UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages

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PART A (80 MARKS)

QUESTION 1

a) Explain two preventive network security mechanisms that control traffic flow or provide remote access.

(2 marks)

b) Explain six strengths of using a Network-Based Intrusion Detection System (NIDS).

(6 marks)

QUESTION 2

a) List two distinct characteristics of the Vigenere cipher as described in the lecture slides.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) State two (2) distinct responsibilities of the Trusted Platform Module (TPM) within the trusted computing architecture.

(2 marks)

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b) Refer to your answer in (a). Justify how the TPM supports the Chain of Trust by detailing six (6) specific aspects of its operation or integration as described in the lecture slides.

(6 marks)

QUESTION 4

a) Explain the two primary motivations for implementing Single Sign-On systems in enterprise environments.

(2 marks)

b) Discuss six distinct characteristics or components associated with biometric authentication methods as presented in the lecture slides.

(6 marks)

QUESTION 5

a) State two key sources or causes of software insecurity as identified in the lecture.

(2 marks)

b) Describe six specific security mechanisms, features, or milestones introduced in Windows operating systems according to the provided timeline and security overview.

(6 marks)

QUESTION 6

a) List two distinct classes of threats to computer security.

(2 marks)

b) Describe six specific password threats identified in authentication security.

(6 marks)

QUESTION 7

a) Explain the primary security focus of the Bell-La Padula model and the Biba model as distinct formal security models.

(2 marks)

b) State six key points regarding access control methods, structures, and goals from the lecture evidence.

(6 marks)

QUESTION 8

a) State TWO (2) types of malware that are described as malicious programs which replicate themselves or exploit system vulnerabilities to cause harm.

(2 marks)

b) State SIX (6) distinct characteristics, impacts, or behaviors associated with malware such as Trojans, worms, or viruses as detailed in the lecture slides.

(6 marks)

QUESTION 9

a) Explain two password security controls used to protect stored credentials and manage access.

(2 marks)

b) Discuss six authentication mechanisms or controls related to Kerberos design, smartcard usage, or password storage.

(6 marks)

QUESTION 10

a) Explain the main concepts of access control by defining a subject and an object, and stating the role of a reference monitor.

(2 marks)

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b) Explain how access control methods implement policies and classify systems into discretionary, mandatory, and role-based categories, including the function of reference monitors.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) Explain the fundamental distinctions between morals, ethics, and laws as frameworks governing human conduct in computer security.

(5 marks)

b) State the defining characteristics of ethics as a framework for professional conduct.

(5 marks)

QUESTION 2

a) Explain how Trojan horses and computer worms differ in their method of replication and the specific harm they cause to host systems.

(5 marks)

b) Discuss the primary operational problem caused by Cross-Site Scripting (XSS) attacks regarding user interaction and browser security boundaries.

(5 marks)

END OF QUESTION PAPER

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